1.4.3 Let $\Lambda = \Lambda_i$. $\wp'$ has a triple pole at 0 and simple zeros at $1/2$, $i/2$, $(1+i)/2$

Since $i\Lambda = \Lambda$ it follows that

$$\wp(iz) = \wp_{i\Lambda}(iz) = i^{-2}\wp(z) = -\wp(z).$$

i) In particular $\wp'((1+i)/2) = 0$, This is a double zero since $\wp'((1+i)/2) = 0$. making it the only zero of $\wp$ as a function on $\mathbb{C}/\Lambda$. Since $\bar{\Lambda} = \Lambda$.

ii) $\wp(\bar{z}) = \overline{\wp(z)}$, so $\wp(1/2) \in \mathbb{R}$, $\wp(i/2) = -\wp(1/2) \in \mathbb{R}$

iii) Compute some dominant terms of $\wp(1/2)$ and $\wp(i/2)$ to show $\wp(1/2)$ is the positive value.

iv) For what m, $\mathbb{C}/m\Lambda \leftrightarrow y^2 = 4x(x-1)(x+1)$?

Sol: i):

$$\wp\left(i\,\tfrac{(1+i)}{2}\right) = \wp\left(\tfrac{(i-1)}{2}\right) \underset{\Lambda}{=} -\wp\left(\tfrac{(1+i)}{2}\right).$$

$$\wp\left(i\,\tfrac{(i-1)}{2}\right) = \wp\left(-\tfrac{(1+i)}{2}\right) \underset{\Lambda}{=} -\wp\left(\tfrac{(i-1)}{2}\right) = \wp\left(\tfrac{(1+i)}{2}\right).$$

By Ex1.4.2 (a), $\wp$ even, $\wp$

Since $\frac{(i-1)}{2} \underset{\Lambda}{=} \frac{(1+i)}{2}$. nonzero half-periods: $\frac{1}{2}, \frac{i}{2}, \frac{(1+i)}{2}$.

hence $\wp\left(\frac{1+i}{2}\right) = \wp\left(\frac{(i-1)}{2}\right) = -\wp\left(\frac{(1+i)}{2}\right)$.

Thus $\wp((1+i)/2) = 0$.

ii): $\wp(\frac{1}{2}) = \overline{\wp(\frac{1}{2})} \Rightarrow \wp(\frac{1}{2}) \in \mathbb{R}$.

$$\wp(\tfrac{i}{2}) = \wp(i \cdot \tfrac{1}{2}) = -\wp(\tfrac{1}{2}) \in \mathbb{R}.$$

iii): $\wp(z) = \frac{1}{z^2} + \sum_{w\in\Lambda}{}' \left(\frac{1}{(z-w)^2} - \frac{1}{w^2}\right) \Rightarrow \wp(\frac{1}{2}) = 4 + \cdots > 0.$

$\wp(\frac{i}{2}) = -4 + \cdots < 0.$

iv): For E, branch point: $0, 1, -1, \infty$,

$e_1 = \wp(\frac{1}{2})$, $e_2 = \wp((1+i)/2) = 0$, $e_3 = (i/2) = -e$,

$\Rightarrow y^2 = 4x(x-e_1)(x+e_1)$. $x \mapsto x/e_1$

$m = \sqrt{e_1} = \sqrt{\wp(1/2)}$.

(1.4.3) Reason similarly with $\Lambda = \Lambda_{\mu_3}$

to find zeros of $\wp$. show $\wp(1/2) \in \mathbb{R}$.

For what m does $\mathbb{C}/m\Lambda$ correspond to

$$y^2 = 4(x-1)(x-\mu_3)(x-\mu_3^2) \ ?$$

Sol: $\mu_3 \Lambda = \Lambda$. $\Rightarrow$ $\wp(\mu_3 z) = \wp_{\mu_3\Lambda}(\mu_3 z) = \mu_3^{-2}\wp(z)$

$= \mu_3 \wp(z)$.

Thus the three half-periods: $1, \mu_3, \mu_3^2$.

$\overline{\Lambda_{\mu_3}} = \overline{\mu_3}\mathbb{Z} \oplus \mathbb{Z} = -\mu_3\mathbb{Z} \oplus \mathbb{Z} = \Lambda_{\mu_3} \Rightarrow \wp(\bar{z}) = \overline{\wp(z)}$.

If $\wp(z) = 0$ then $\wp(\mu_3 z) = \mu_3 \wp(z) = 0$,

so $\mu_3 z \equiv \pm z \ (\Lambda)$. since $\wp$ even and has only 2 zeros

Thus $\mu_3 z \equiv z \ (\Lambda)$. (if take $-z$, then $(1+\mu_3)z \in \Lambda$.

$\Rightarrow$ $-\mu_3^2 z \in \Lambda \Rightarrow z \in \frac{1}{2}\Lambda$.)

Thus $z = (1+2\mu_3)/3$, $(2+\mu_3)/3 \equiv -(1+2\mu_3)/3 \ (\Lambda)$.

Since $(\mu_3 - 1)z = -\mu_3^2(1+2\mu_3)/3 \in \Lambda$.